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Stabilization of Crude Oil Emulsions by Flowback Fluid Solids Post Well Drilling and Workover Activities

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SUMMARY

In order to increase efficiency and to align with the United Nations' Sustainable Development Goals (SDGs) to cut down on greenhouse gas (GHG) emissions and achieve net zero, flowback fluids from drilling operations are sent to the production line and not to rig burners for treatment with zero flaring. However, due to different nature of substances present in the flowback fluid, crude oil emulsions can be stabilized by the flowback fluid leading to a difficult and complex oil and water separation process. In this paper, a study to find the root cause of poor oil-water separation and process upset due to crude oil emulsion stabilization by flowback fluids was presented. Advanced analytical techniques including rheo-microscopy, high performance liquid chromatography (HPLC), Fourier-transform ion cyclotron resonance mass spectrometry (FT-ICR MS), Fourier-transform infrared (FT-IR) spectroscopy and X-ray diffraction (XRD) were deployed to better understand the nature, behavior and molecular-level characteristics of the crude oil emulsion during flowback activities. Crude oil emulsion sample collected at the beginning of the process upset showed elevated viscosity and pour point, along with a high asphaltene content, indicative of increased heavy and interfacial molecular compounds as shown by petroleomics analysis. XRD analysis identified higher concentrations of magnetite (Fe_3O_2) and iron (Fe) particles in the early-stage sample when compared with the samples obtained at latter stage of the process upset. The study confirmed that asphaltenes, magnetite and iron particles from flowback fluid, synergistically stabilized the emulsion by forming a rigid interfacial film that hinders water droplet coalescence. The research suggests that rapid compatibility and emulsion separation testing of flowback fluids after workover activities can prevent similar process upsets. The findings also highlighted the need for ensuring corrosion products and other solid particles are completely removed from the drilling and workover fluids using appropriate technologies for successful flowback operations. The research will enhance the understanding of emulsion dynamics and offer actionable insights for optimizing separation processes in the oil industry and for enhanced total flowback to production lines with zero flaring for reducing carbon emissions from drilling and workover operations.

INTRODUCTION

After well maintenance operations are conducted, for example, acidification to remove clogging near wells for control of formation damage, cleaning of the wellbore for corrosion control, or replacement of a damaged or worn-out equipment, the fluid from the bottom of the well (known as flowback fluid) is flowed back from the bottom of the well to the surface for treatment and subsequent production of the well (Suleimanov et al. 2018; Bai et al. 2021). This flowback fluid contains large amount of oil and water mixture, solids substances (e.g., scaling and corrosion products), and residual acids, polymers and surfactants (Li et al. 2022).

Conventionally, flowback fluids from offshore and onshore well clean-up, testing and workover operations are channeled to the rig burners for burning of hydrocarbons. Burning of hydrocarbons using rig burners is one of the sources of hydrocarbon emissions (unburnt hydrocarbons, toxic gasses, and soot) into the atmosphere during oil well drilling and workover operations (Hassan et al. 2001; Ugo' and Horace 2017). As part of oil and gas industry efforts to increase efficiency and to align with the United Nations' Sustainable Development Goals (SDGs) of cutting greenhouse gas (GHG) emissions and achieving net zero (Mire et al. 2018), flowback fluids from drilling operations are sent to the production line for treatment with zero flaring.

As pointed out by different studies, due to different concentrations of organic and inorganic substances present in the flowback fluid, crude oil emulsions can be stabilized when flowback fluid is channeled back to the production line leading to a difficult and complex oil and water separation process (Bai et al. 2021). In their study, Shirazi et al. (2019) quantitatively evaluated acid-oil emulsion and asphaltic sludge formation following acidization treatment of near wellbore region. They showed that the acid-mixture ratio, acid concentration and presence of ferric ions were the main factors affecting the emulsion stability and the amount of sludge formed when crude oil is exposed to the acid. Li et al. (2022) studied the effects of various components (surfactants, polymers, inorganic solids, solid particles and asphaltenes) on the stability of emulsion from flowback fluids after acid cleaning and descaling treatment of oil wells in the DaQing oilfield of China. They found that NaCl charges, silica (SiO_2) nanoparticles, FeS nanoparticles, and CaCO_3 nanoparticles facilitated a multi-factor synergistic emulsification of the emulsion through forming a dense wrapping layer between oil and water droplets which could interfere with flocculation and coalescence and subsequent separation of water droplets from crude oil emulsions. When asphaltene concentration is higher than 0.67%, the stability of the emulsion increases and emulsion separation becomes more difficult.

Abdul-Ameer et al. (2006) showed that when acidification is applied on the well pipe strings, ferrous ions are formed due to metal pipe corrosion which oxidize to ferric ions in the surface treatment facilities. These ferric compounds from the flowback fluid can stabilize crude oil emulsion and make separation process difficult. In their water - model oil emulsion experiment, Zheng et al. (2022) found that stable water-in-oil model oil and water-in-oil crude oil emulsions were formed with increase in Fe (III) species. Additionally, addition of ethylene oxide was shown to improve the stability of the water-in-oil emulsion in the presence of iron (III) species.

Apart from the naturally available interfacially active components in crude oils that are known to stabilize emulsion, such as asphaltenes and resins, presence of other solids and the nature of their particles can determine the type and stability of the emulsion that will be formed. These solids can include clays, crystalline waxes, and corrosion and scaling products. Stabilization of emulsions by solid particles based on adsorption and formation of a dense layer at the interface of oil and water droplets thereby preventing the droplets from coalescence is known as pickering stabilization (Umar et al. 2018).

It was suggested by Kilpatrick (2012) that the particle-stabilized emulsion (pickering emulsion) is stabilized through enhancement of crude oil acid-stabilized emulsions or asphaltenes/resin stabilized emulsions when surface-modifying components in the crude oil adsorb the solid particles that facilitate their interfacial activities. These solid particles are more effective in stabilizing emulsions when they are in tens to hundreds of nanometer size scale. In their experiment, Kokal and Al-Juraid (1998) have shown

that for solid particles (inorganic scales) to stabilize emulsions, they need to be coated with heavy polar components of crude oil such as asphaltenes which will make the solids intermediately-wet. This will allow the solids to stay at the oil - water interface and prevent the droplets from coalescence and therefore make the emulsion tight.

Sullivan and Kilpatrick (2002) studied in detail how inorganic solids (clay kaolin, iron oxide, and clay montillonite) interact with asphaltenes and resins to stabilize emulsions. They showed that inorganic solid particles can stabilize emulsions more when they are present in small sizes (few microns) and in significant concentrations. Similarly, stabilization of emulsions by solids is enhanced by wettability modification of solids to be preferentially oil-wet due to adsorption by asphaltenes and resins in the crude oil. Additionally, they highlighted the importance of asphaltene state of aggregation which is controlled by crude oil medium solvency (aliphatic/aromatic ratio and resin concentration) which determines solubility of asphaltene aggregates for effective stabilization of solid particles.

It is almost impossible to totally eliminate emulsions during crude oil production, and emulsion treatment and prevention are known to account for huge part of the operating budget in wet crude processing plants (Kokal et al. 2007; Ismail et al. 2020). The drain on operating budget from emulsion treatment cost becomes very high and evident when handling tight crude oil emulsions. This emulsion tightness increases due to presence of stabilizing solid substances arising from inorganic scales and corrosion products in reservoir fluids and from flowback fluids from well drilling and workover activities leading to high consumption of demulsifier, oil-water poor separation and process upset.

Herein, a study on stabilization of crude oil emulsions by flowback fluid solids following well drilling and workover activities was presented. Flowback fluids received by one of the wet crude handling facilities (WCHFs) in Saudi Arabia resulted in poor oil-water separation, process upset and difficulty in crude oil separation process. Despite increasing demulsifier dosage by about 37 percent increase, no clear oil-water separation was achieved during and after the flowback activities were halted. The poor oil-water separation lasted for weeks after which the normal and clear separation was restored.

To study and investigate the root cause of the crude oil emulsion stabilization and tightness stemming from flowback activities after workover operations, a sample of the crude oil emulsion (sample A) at the inlet of the WCHF was collected immediately when process upset was observed during flowback operations. This served as the emulsion sample at the initial stage of the process upset (early-stage sample). Two additional emulsion samples (sample B and sample C) were then collected after some days when poor oil-water separation started to reduce indicating last stage of the process upset (latter-stage samples). All the three emulsion samples were collected at the inlet of the WCHF before injection of chemicals and characterized using different analytical techniques including fundamental laboratory tests and advanced analytical techniques for molecular-level analyses.

EXPERIMENTAL SECTION

Emulsion Water Separation, Visual and Microscopic Properties

A portion homogenized emulsion sample was transferred from a metal tin container to a glass bottle and visually observed for the presence of free water, precipitate, solids and sediment. Another portion of the samples was then transferred to a centrifuge tube without adding demulsifier and heated in a water bath at 150°F for 15 minutes to observe water separation behavior of the sample.

An emulsion separation test was then used to study water separation and evaluate the tightness (stability) of the emulsion sample. The test is known as the emulsion separation index (ESI) and it is defined as follows:

$$ESI = \frac{\Sigma \text{Water separated at a given demulsifier concentration and time (\%)}}{\text{Number of recordings}}$$

Where the water separation is in the percentage of basic sediment and water (BS&W) value and ESI ranges from zero (no separation) to 100 percent (full separation). Thus, the separation index provides a measure of emulsion tightness and the lower the index, the tighter the emulsion. The index is quoted at the temperature of the test and the demulsifier dosage. This procedure was described in detail by Kokal et al. (2007).

To study the inner structures and nature of the phases in the emulsion sample, rheo-microscopic imaging of the emulsion sample was carried out using Dynamic Mechanical Analyzer (MCR 702 Multidrive Rheometer) from Anton Paar. The analyzer was integrated with a polarized light microscopy imaging system. The imaging uses a 43 mm diameter parallel plate measurement system of transparent quartz glass. A microscope, high-intensity 150 W LED light source, and regulator are integrated with the lower part of the stage. The light source illuminates the sample vertically from below when the rotor drives the sample on the transparent glass. Images of the microscopic morphology and dynamic behavior of the sample can be collected by the microscope in real time. The microscope has an objective lens of 20x magnification with a focal length of 30.9 mm and a lens depth of 1.6 μm with optical compensation. The microscope can achieve a resolution of 0.7 μm . The field of view is 440 $\mu\text{m} \times 330 \mu\text{m}$.

Physical Properties of Emulsion Samples

Basic sediment and water (BS&W) were measured according to ASTM D4007. This method was applied to determine water and sediment content in the emulsion samples. 50 ml each of crude oil emulsion sample and water-saturated toluene were placed into a cone-shaped centrifuge tube. The mixture was shaken and then placed in a water bath at 140°C for 15 minutes. After centrifugation, the volume of the higher density water and sediment layer at the bottom of the tube was read and noted. The API gravity of the prepared crude samples was determined using ASTM D4052 method. The test was conducted at 60°F for the determination of API gravity. The pour point of the samples was determined using ASTM D97 method to determine the temperature at which the crude oil begins to gel.

The dynamic viscosity of the emulsion samples was measured using Anton Paar MCR-702 rheometer. A double gap concentric cylinder was used with a cup geometry. The viscosity was measured using a temperature ramp up of 50 to 158°F at a constant shear rate of 100 s^{-1} .

Compositional Properties of the Emulsion Samples

Distribution of hydrocarbons (saturates and aromatics) was measured using high performance liquid chromatography (HPLC) in combination with a refractive index detector to determine the distribution of hydrocarbons in emulsion samples. The distributions of hydrocarbons in terms of saturates, 1-ring aromatics, 2-ring aromatics and 3+-ring aromatics were determined. The HPLC used in this method consists of a micro vacuum degasser, a binary pump, a column thermostat, an auto-sampler and a refractive index detector. The HPLC conditions set for the method are flow rate of 0.8 mL/min, refractive index detector, Injection volume of 10 μL , run time of 30 minutes, mobile phase used was n-heptane with LiChrospher NH_2 (250 $\times$ 4.6 mm) column and column oven temperature of 35°C.

Fourier Transform Ion Cyclotron Resonance Mass Spectrometer (FT-ICR MS) was used study aromatic compounds class distributions in the range of C_{10} to C_{80} containing aromatic rings of 1 to 9+. The sample was extracted with toluene and diluted before the analysis. The mass spectrometry characterization for the samples was acquired using a FT-ICR MS operated with an atmospheric pressure photoionization (APPI) ion source in a positive mode. A syringe was used to infuse the sample at a flow rate 20 $\mu\text{L}/\text{min}$. The drying temperature was set to 400°C. As a result, the mass spectra were acquired in the 150 – 3000 amu mass range.

Geochemical Properties of Produced Water

Geochemical properties of separated water from emulsion samples were measured using different methods. Anions in water were measured using chemically suppressed ion chromatography (ASTM D4327),

measurement of pH in water was obtained using ASTM D 1293, while metals in water were measured using inductively coupled plasma atomic emission spectroscopy (ICP-AES).

Determination of emulsion stabilizing solids

Presence and concentration of emulsion stabilizing solids such as corrosion and scaling products and asphaltenes were measured using X-ray powder diffraction (XRD), Fourier transformed infrared spectrometer (FTIR), and FT-ICR MS.

For determination of corrosion and scaling products using XRD analysis, a portion of the emulsion sludge was dissolved in toluene to remove most of the hydrocarbon components and then 0.45 μm was used to filter the solids components and allowed to dry. The filtered samples were then mounted into the sample holder of the PANalytical Aeris X-ray powder diffractometer operated with a Cobalt X-ray tube. XRD data were collected from 5 to 90° at 2 θ Bragg angles with a step size of 0.04° and counting time of 16 minutes. Phase identification was performed considering the match of the ICDD PDF4+ database references with the measured XRD data over the whole measured range. The Rietveld method, implemented in the same software, was employed for the quantitative analysis.

Extraction followed by centrifugation were used repeatedly to extract asphaltenes with n-heptane using a 100mL centrifuge tubes (El Nagy et al. 2022). After several cycles of asphaltene extraction with n-heptane and centrifugation, toluene was used to dissolve the extracted asphaltenes and then evaporated by heating at 110°C. The extracted asphaltenes in the three crude emulsion samples were then characterized by FTIR and FT-ICR MS techniques. FTIR spectroscopic data was acquired for the extracted asphaltene samples using a Nicolet iS50 FT-IR spectrophotometer equipped with deuterated triglycine sulfate (DTGS) detector and Specac Golden gate diamond ATR accessory with an average of 128 scans at a resolution of 4 cm^{-1} .

Preparation of Emulsion Samples for Analyses

Based on the nature of the crude oil emulsion samples, each of the three emulsion samples was apportioned into four as described below for further analytical tests to determine its properties:

- i. The crude oil emulsion sample as-received, before separation or addition of demulsifier, was used for determination of BS&W, ESI, dynamic viscosity, and FT-ICR MS analysis.
- ii. The bottom layer (sludge) obtained after leaving the sample to settle at room temperature (24°C) for 24 hours followed by removal of the top oil layer. This sample was used for XRD analyses.
- iii. The top oil layer obtained after removing water using a demulsifier and heating. This sample was used for determination of density, API, asphaltenes, and detailed hydrocarbon analysis.
- iv. The separated water obtained after removal of the top oil layer using demulsifier and heating was used for geochemical analysis.

RESULTS AND DISCUSSION

Visual Observations, Water separation and Microscopy of the Emulsion Samples

Emulsion Visual Observations. A portion of the homogenized sample A was transferred from a metal tin container to a glass bottle and visually observed. There was no free water and no solids were observed. However, a bottom emulsion layer was observed when the sample stood for hours at room temperature (24°C) (Figure 1a). Then a portion of the sample was transferred to a centrifuge tube and heated without adding demulsifier in a water bath at 150°F for 15 minutes. No water separation was observed (Figure 1b). Water separation was only observed after adding an excess amount of demulsifier (>500ppm) and heating at 150°F for 15 minutes (Figure 1c). The separated water appeared to be dirty but there was no sediment or solids observed in the separated water at the bottom of the tube.

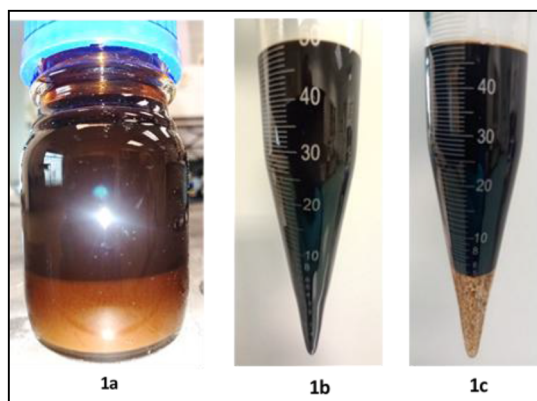


Figure 1—Emulsion samples showing bottom emulsion layer (1a); no water separation after heating at 150°F for 15 minutes (1b); and separated water bottom layer after adding an excess of demulsifier and heating at 150°F for 15 minutes (1c).

Emulsion Water Separation and Tightness. The water separation and tightness of the emulsion samples were measured using emulsion separation index (ESI). A high value of ESI indicates loose emulsions, while a lower value indicates tight emulsions. The test was conducted on the samples to check the tightness of the emulsions. Experiments were performed at room temperature (75°F) and at 150°F using incumbent demulsifier chemical. There was no water separation observed from experiments carried out at room temperature. Fig. 2 showed the water separation behavior and ESI values of the sample A crude oil emulsion. Significant water separation was only observed using 200 ppm demulsifier dosage at 150°F. The water separated was dirty in all the ESI experiments carried out. The dirty water as seen in Figure 1c was an indication of poor separation and oil carryover as observed in the field from the screen of separating vessel. The ESI value of 34% observed at 150°F and 200 ppm demulsifier dosage was an indication that the emulsion sample was tight and difficult to separate.

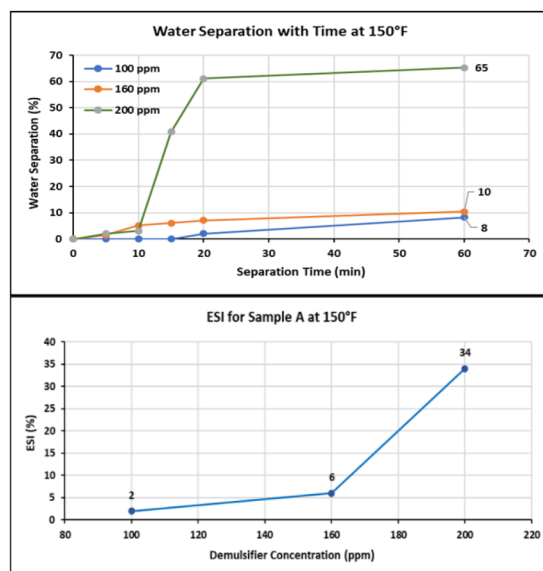


Figure 2—Water separation behavior (left) and ESI values (right) for sample A.

Emulsion Rheo-microscopic Imaging. The images of the emulsion sample obtained from rheo-microscopy are presented in Figure 3. The emulsion contains majorly very small water droplets dispersed in the continuous oil phase. Observed also were dark solid particles dispersed and distributed within the water and oil droplets of the emulsion. Some of the solid particles were smaller while some were larger than the water droplets in the emulsion. Solid particles are known to be adsorbed at the oil-water interface and form

a protective film to prevent water droplets from coalescence. This happens particularly when the solids particle sizes are smaller than the sizes of the emulsion droplets and when the solids are intermediately-wet by the water and oil droplets when coated by the heavy polar components of crude oil (e.g., asphaltenes) (Kokal and Al-Juraied 1998; Sullivan and Kilpatrick 2002; Kilpatrick 2012).

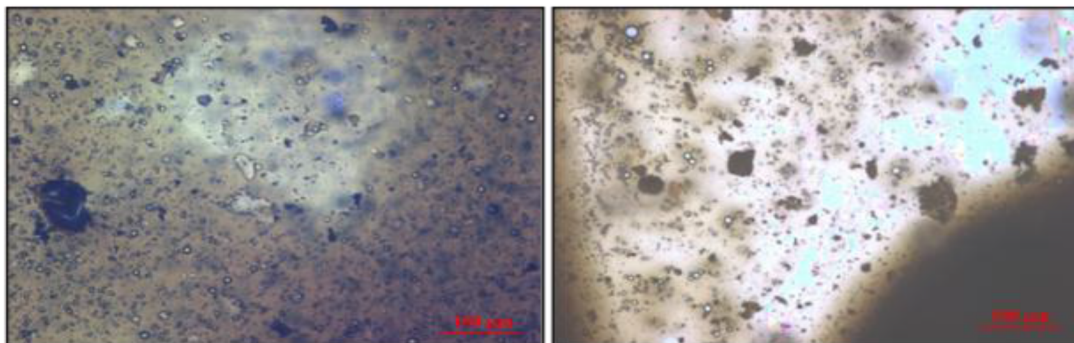


Figure 3—Rheo-microscopic images of crude oil emulsion (sample A) showing nature and distribution of water droplets and solid particles within emulsion droplets.

From the results obtained of visual observations, water separation and microscopy of the emulsion sample, it was deduced that the emulsion sample (sample A) obtained at the beginning of the flowback operation was tight and difficult to separate unless when high dosage of demulsifier is applied at high temperature of 150°F. The emulsion sample also contained solid particles of various sizes distributed within the emulsion droplets which can prevent water droplets from coalescence leading to difficulty in emulsion separation.

In order to identify the presence, nature and concentration of substances coming from flowback fluids (e.g., scaling and corrosion products, asphaltenes, clay particles, etc.) that resulted in stabilizing crude oil emulsions leading to process upset, the emulsion properties of sample A collected at the beginning of the flowback operations (at the early stage of the process upset) were determined and compared with the emulsion properties of the samples B and C collected at the last stage of the process upset. Therefore, in the following sections, properties of the three emulsion samples (A, B and C) were identified and compared in relation to emulsion-stabilizing organic and inorganic solids.

Properties of the Emulsion Samples

Physical Properties. Results obtained for physical properties of the three emulsion samples have been presented in Table 1, while dynamic viscosity curves were depicted in figure 4. A gradual decrease in the values of density, pour point and viscosity was observed from sample A to sample C. This shows that the sample A collected at the early stage of the process upset contained heavy substances (organic and/or inorganic compounds) that were brought into the main production line along with the flowback fluids. Since flowback operation was halted immediately after process upset was observed, the concentration of these heavy compounds would be higher at the beginning of the process upset when sample A was collected than towards the end of the process upset when samples B and C were collected.

Table 1—Physical properties of the crude oil emulsion samples

Physical Properties	Samples		
	A	B	C
API Gravity @ 60°F	31.50	31.70	32.30
Density @ 60°F (g/cm ³)	0.8676	0.8662	0.8637
Pour Point (°F)	0	<-20	<-20
BS&W (Vol %)	9.8	8.0	7.6
Viscosity (mPa.s) *	7.01	5.92	4.56

* Viscosity was measured at 150°F and shear rate of 100 S⁻¹

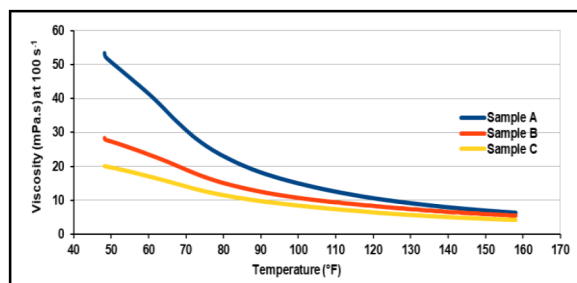


Figure 4—Dynamic viscosity of the emulsion samples using a temperature ramp of 50 to 158°F at a constant shear rate of 100 S⁻¹.

The measurements of dynamic viscosity were conducted using a temperature ramp of 50 to 158°F at a constant shear rate of 100 S⁻¹. These values have shown that sample A was more viscous (7.01 mPa.s) than the sample B and C (5.92 and 4.56 mPa.s, respectively). Viscosity of emulsions depends on a number of factors including viscosity of oil and water phases, distribution of droplet sizes, amount of dispersed water in the emulsion, temperature, shear rate, and amount of solids present (Kokal 2005). The high viscosity values observed for Sample A when compared to Samples B and C could be related to the relatively high dispersed water content and solids (organic and inorganic solids) from flowback fluids.

Distribution of Hydrocarbons by HPLC. Results for the distribution of hydrocarbons in the three samples were presented in Table 2 and HPLC overlaid liquid chromatograms were depicted in Figure 5. The HPLC technique was used to determine distributions of hydrocarbons in terms of saturates, 1-ring aromatics, 2-ring aromatics and 3+-ring aromatics for comparison purpose among the three emulsion samples. Though the distribution of hydrocarbons obtained using this method was not absolute due to the unknown response of refractive index detection; however, it was used among a set of samples for a comparison purpose. The HPLC analysis has shown that the distribution of hydrocarbons was similar for the three emulsion samples with distributions of saturates slightly decreasing from sample A (43.2%) to samples B (42.0%) and C (41.3%). Conversely, distribution of aromatics appeared to be slightly increasing from sample A to samples B and C.

Table 2—Distribution of hydrocarbons (saturates and aromatics) by HPLC

Hydrocarbon Distributions	Samples		
	A	B	C
Saturates (%)	43.2	42.0	41.3
1-ring (%)	24.5	25.3	25.7
2-ring (%)	17.3	17.9	17.9
3+-ring (%)	15.0	14.9	15.1

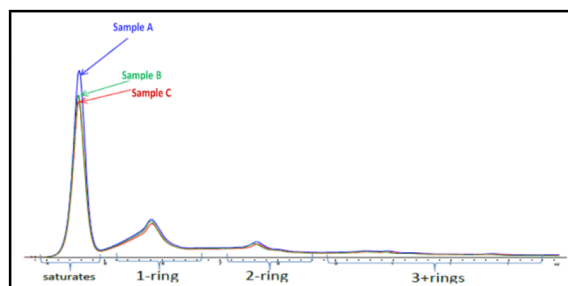


Figure 5—Overlaid liquid chromatograms of emulsion samples (Sample A, B & C) by HPLC analysis

Aromatics and NSO Compounds Class Distributions by FT-ICR MS. Results from petroleomics analysis (application of FT-ICR MS to study chemical compositions of petroleum at the molecular level) have been displayed in Figure 6. The analysis has revealed that the aromatic portion of the three emulsion samples were almost identical and contained mainly aromatic hydrocarbon compounds (HC), sulfur-containing compounds (mono-(S₁), di-(S₂) and tri-(S₃)), nitrogen and oxygen containing compounds (Figure 6). The distribution of these compounds is in the range of C₁₀ – C₇₀₊ with aromaticity of 1 – 9 aromatic rings. The results showed that the aromatic content of the three emulsion samples were typical of crude oil samples. The analysis revealed nine types of compounds in the emulsion samples including aromatic hydrocarbons (HC) and NSO heteroatom compounds (S₁, S₂, S₃, N₁, N₁S₁, O₁, and O₁S₁). Similar abundances were found for the HC, S₁, S₂, S₃, N₁, N₁S₁, and O₁ species in the three emulsion samples. O₂ class of compounds were observed mainly in sample A and C, while O₁S₁ compounds were found with high abundance only in sample A and little or no presence in samples B and C. Presence of O₁S₁ (sulfoxides) in emulsion sample A is an indication of presence of heavy hydrocarbon components when compared with samples B and C. This is because relative abundance of O₁S₁ species, increases with increase in heavy and interfacial materials that stabilizes water-in-crude oil emulsions (Ballard et al. 2020; Ji et al. 2021).

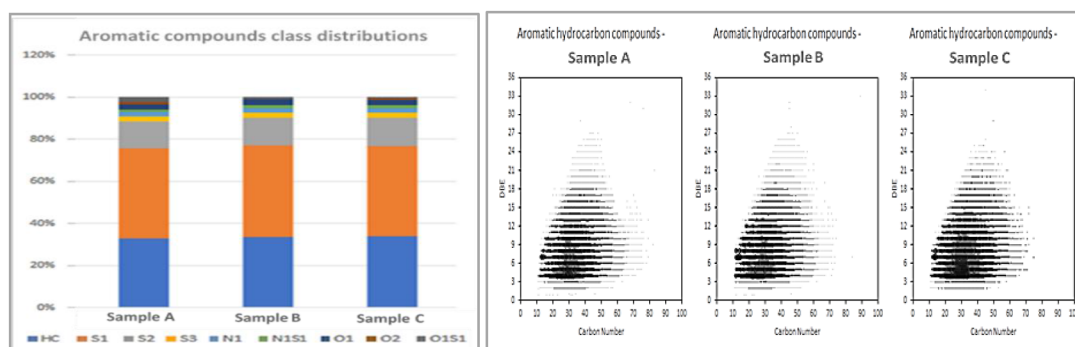


Figure 6—APPI (+) FT-ICR MS relative abundance (%) of aromatic and heteroatom classes (left) and plots of DBE vs. carbon number of aromatic compounds (right) in the three emulsion samples.

Determination of Asphaltenes in the Emulsion Samples. To further understand the presence and nature of polar and interfacial materials in the three emulsion samples, the concentration of asphaltenes were determined by the centrifugation method using n-heptane as a solvent (El Nagy et al. 2022). Results obtained for measurement of asphaltenes have been shown in Table 3, where sample A showed higher concentration of asphaltenes (2.22 Wt. %) when compared to sample B (1.77 Wt. %) and sample C (1.93 Wt.%). The functional groups of the n-heptane extracted asphaltenes were further confirmed using FTIR and characterized at molecular level using FT-ICR MS technique.

Table 3—Asphaltene content of the three emulsion samples

Asphaltenes Content	Samples		
	A	B	C
Asphaltenes (wt. %)	2.22	1.77	1.93

The acquired data from the FT-IR scan for the three n-heptane extracted samples were almost identical with slight differences in the signal intensities at the fingerprint region (Figure 7). The spectra showed that the asphaltenes contained a significant number of aromatics ($C=C$ stretching at 1598 cm^{-1}). The presence of sulfoxide functional group ($C_2S=O$) was indicated at 1031 cm^{-1} . These sulfoxide functional groups were considered to be O_1S_1 class species observed from the FT-ICR MS analysis. Absorption at 729 cm^{-1} was related to alkyl chains longer than four methylene ($-CH_2$) units. These spectra of asphaltenes were similar to FTIR spectra of asphaltenes reported in other studies (Kokal and Al-Juraid 1998; Peksoz et al. 2011).

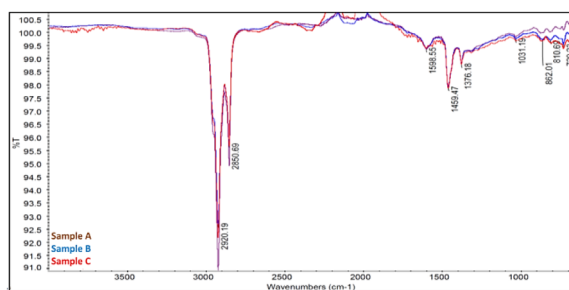


Figure 7—FTIR Spectra of n-heptane extracted asphaltenes for the three emulsion samples.

Additionally, the FT-ICR MS analysis of the extracted asphaltenes has shown that, like in any other asphaltenes, pure aromatic hydrocarbons, heteroatomic aromatic compounds of sulfur, nitrogen and oxygen-containing compounds were present in the asphaltenes of the three emulsion samples (Figure 8). The data revealed aromatic and NSO heteroatomic compounds in the samples comprising of HC , S_1 , S_2 , S_3 , S_4 , N_1S_1 , N_1S_2 , N_1S_3 , N_1 , N_2 , O_1 , O_2 , O_3 , O_1S_1 , O_1S_2 , and O_1S_3 . Different relative abundance of these compounds was found in sample A when compared to samples B and C. Samples B and C showed similar relative abundances of the aromatic and NSO compounds. Sample A was found to be enriched in N_xS_y (N_1S_1 , N_1S_2 and N_1S_3) and O_xS_y (O_1S_1 , O_1S_2 , and O_1S_3) heteroatoms while samples B and C were enriched in HC and O_x (O_1 , O_2 and O_3) classes of compounds. In examining plots of DBE versus carbon number, the diagonal band of data points observed at higher DBE is known as the asphaltene range, while the lower DBE region represent the maltenes – remaining oil components. What was obvious in the DBE versus carbon number plots of the three samples was that, sample A contained more asphaltenes when compared to samples B and C. This finding agreed with the concentrations of asphaltenes obtained using centrifugation method in the three emulsion samples with sample A having the highest asphaltenes concentration.

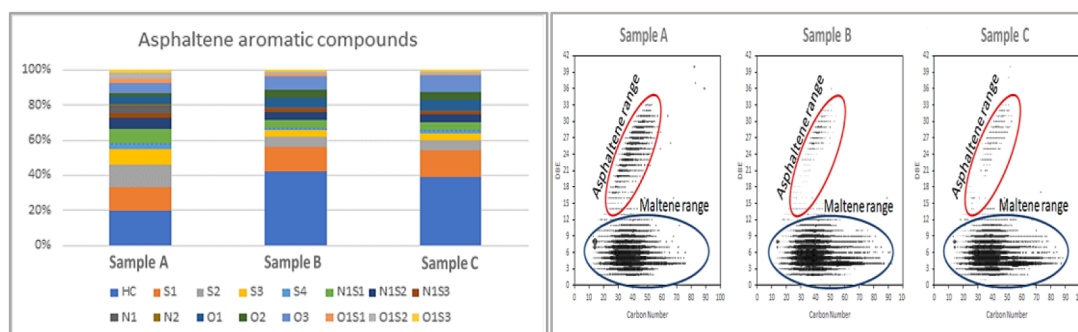


Figure 8—APPI (+) FT-ICR MS relative abundance (%) of aromatic and heteroatom classes (right) and plots of DBE vs. carbon number of aromatic compounds (left) in the n-heptane extracted asphaltenes.

X-ray Powder Diffraction (XRD) of Emulsion Samples

In order to identify and characterize the compositions of the solid particles observed in the emulsion droplets from rheo-microscopic imaging, XRD analysis was conducted. The XRD patterns and their corresponding phase identifications for the three samples are shown in Figure 9. Quantitative phase analysis results are depicted in Table 4. Halite (NaCl) was present in all the three emulsion samples, whereas Magnetite (Fe_3O_4) was observed only in samples A and C. Iron (Fe) with large grains was also found in the sample A. The Magnetite found in sample A (12.1 wt.%) and sample C (5.9 wt.%) was expected to be due to the wet corrosion of metallic pipes and tubing in the process. The iron (most likely from steel) was thought to be metal shavings from metal scrapping of pipes during well workover and completion operations. These inorganic solids were introduced into the inlet of WCHF by the flowback fluids. Therefore, results from XRD analysis indicated that sample A collected at the beginning of the process upset contained higher concentrations of these solids (iron and magnetite) than the samples B and C which were collected at the latter stage of the process upset. Solids such as corrosion products (magnetite and metal particles in this case) stabilize emulsions by forming a "skin" on water droplets and prevent the dispersed water droplets from coalescing and make the oil-water separation difficult (Shirazi et al. 2019).

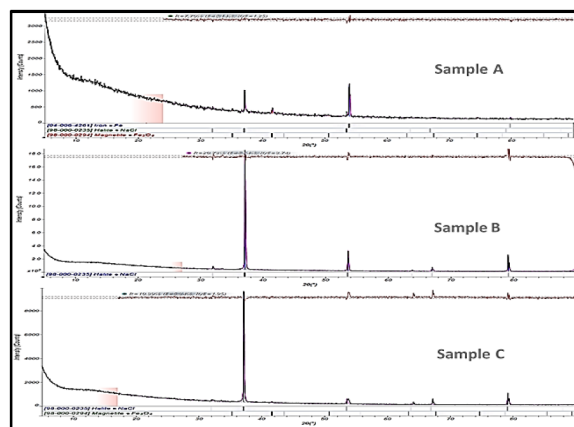


Figure 9—X-ray diffraction pattern of the emulsion samples with their corresponding phase identifications

Table 4—Quantitative phase analysis of the emulsion samples

Phases	Samples		
	A	B	C
Iron (Fe), wt%	56.3	-	-
Magnetite (Fe ₃ O ₄), wt%	12.1	-	5.9
Halite (NaCl), wt%	31.6	100.0	94.1

Geochemical Properties of Separated Produced Water

In order to understand variations in geochemical properties of produced water separated from the three emulsion samples, geochemical analysis was conducted and the data obtained for the three samples are presented in Table 5. The results showed that geochemical properties of the separated water from the three emulsion samples were similar. The concentrations of most of the metals in the water appeared to be lower in sample A than in samples B and C, except for chloride and strontium concentrations. This could be related to the increase in volume of brine from flowback fluids at the beginning of the process upset. pH values of 4.4 – 4.7 were obtained from the separated water samples. Low pH values were shown to make interfacial film of the emulsion rigid thereby making water separation difficult (Kokal 2005). Additionally, iron and manganese were found to be present in the separated water from the three emulsion emulsions. This could be related to the iron and magnetite phases identified by XRD analysis which could be due to wet corrosion products introduced into the WCHF by the flowback fluids. The values of iron concentrations in the separated water samples (e.g., 19 mg/L for sample A) appeared to be low when compared to the concentrations of iron and magnetite phases identified by XRD analysis (56.3 wt.% iron and 12.1 wt. % magnetite for sample A). This phenomenon could be due to the fact that the solids are oil-wetted by the heavy interfacial materials in the emulsion such as asphaltenes. It is therefore, expected that iron particles will align more in the emulsion oil phase than in the separated water.

Table 5—Geochemical properties of produced water separated from emulsion samples

Properties	Samples		
	A	B	C
Barium, mg/L	7	6	6
Calcium, mg/L	14,732	15,197	17,220
Chloride, mg/L	159,126	150,826	155,144
Conductivity, $\mu\text{S}/\text{cm}$ @ 25 °C	>10,000	>10,000	>10,000
Total Dissolved Solids, mg/L	240,641	232,591	250,316
Magnesium, mg/L	2,132	2,362	2,353
pH @ 25 °C	4.4	4.7	4.5
Potassium, mg/L	2,362	2,635	3,025
Sodium, mg/L	61,126	60,555	71,448
Specific Gravity @ 60 °F	1.1731	1.167	1.1712
Strontium, mg/L	703	532	590
Sulfate, mg/L	453	478	530
Iron, mg/L	19	23	24
Manganese, mg/L	6	5	5

CONCLUSIONS

Root cause of poor oil-water separation and process upset due to crude oil emulsion stabilization by flowback activities after workover operations have been studied and the following conclusions were made:

1. A stabilized crude oil emulsion was formed by solids stemming from flowback fluids after workover operations. The tight emulsion was separated only after adding an excess amount of demulsifier and heating at high temperature. The emulsion contained small water droplets and dark solid particles of various sizes dispersed in the continuous oil phase.
2. The emulsion sample collected at the beginning of the process upset contained heavy substances from flowback fluids as indicated by its physical properties (e.g., density, pour point and viscosity) when compared to the emulsion samples collected at the end of the process upset.
3. A slight increase in the distribution of aromatic compounds (1-ring, 2-ring, and 3+-ring aromatics) and slight decrease in saturates were observed from emulsion sample collected at the beginning of the process upset when compared to the emulsion samples collected at the latter stage of the process upset. subsequently, a higher asphaltenes content of 2.22 wt.% was obtained in early-stage emulsion sample when compared to asphaltenes contents of 1.93 to 1.77 wt. % found in the latter-stage samples.
4. Petroleomics analysis of the emulsion samples and n-heptane extracted asphaltenes revealed presence of high abundance of N_xS_y (N_1S_1 , N_1S_2 and N_1S_3) and O_xS_y (O_1S_1 , O_1S_2 , and O_1S_3) heteroatom compounds in the early-stage emulsion sample when compared with the latter-stage emulsion samples. Sulfoxides (O_1S_1) are interfacially active materials in asphaltenes that stabilize water-in-oil emulsions.
5. Compositional analysis of solids in emulsion samples showed presence of higher solids content (iron and magnetite) in the early-stage emulsion sample than in the samples collected at the latter stage of the process upset. These inorganic solids were expected to be products of wet corrosion and metal scrapings from well workover operations that were produced together with flowback fluids.
6. From the aforementioned, the root cause of the poor oil-water separation and process upset following flowback operations was found to be due to adsorption of inorganic solids (iron and magnetite particles) by the interfacially active asphaltenes in the flowback fluids. As a result, the asphaltenes, magnetite and iron particles from flowback fluids, synergistically stabilized the emulsion by forming a rigid interfacial film which hinders coalescence of emulsion water droplets leading to poor oil-water separation and process upset.

RECOMMENDATIONS

In order to have a successful flowback operations to the production line after a drilling or well workover operation, the following are some of the recommendations:

1. For well completion and maintenance operations that introduce solid particles into the well (e.g., acidification for control of formation damage, cleaning of the wellbore for corrosion control, or replacement of a damaged or worn-out equipment), there is need for total removal of the solid particles (e.g., scales, corrosion products and metal shavings) from the well to prevent solids from mixing with flowback fluids.
2. Before commencement of the flowback operations to the production line, it is good to conduct a quick compatibility and emulsion separation test for the incoming flowback fluid and the existing fluid in the main production line. This will indicate possible emulsion separation problems that may arise due to incompatibility of the two fluids.
3. When flowback fluids are expected to cause emulsion separation problems, stand-alone treatment facilities or where possible, test separators may be utilized to treat flowback fluids before channeling to the main treatment facilities.

4. Where flowback fluids are expected to cause separation problems due to solids, effective chemical demulsifiers that can enhance separation of solids-stabilized emulsions can be utilized in order to prevent poor oil-water separation in the wet crude handling facilities.

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